

College Admission Agent (RAG Based)

Problem Statement No. 4

Project Title:-

Admit_AI – AI-Powered Multi-College Admission Assistant using IBM Granite & Retrieval-Augmented Generation (RAG)

Problem Overview

Choosing the right college and understanding admission procedures is one of the biggest challenges for students. Every university publishes admission information in different formats such as brochures, fee structure PDFs, hostel documents, placement reports, and eligibility guidelines.

Students often spend hours searching multiple websites to find accurate information regarding:

- Admission eligibility
- Course details
- Fee structure
- Hostel fees
- Scholarship
- Placement statistics
- Required documents
- Admission process
- Important deadlines

Since this information is scattered across different documents, students frequently receive outdated or incorrect information.

An intelligent AI-powered admission assistant can simplify this entire process by providing instant answers from official college documents.

Problem Statement

Develop an AI-powered College Admission Agent using Retrieval-Augmented Generation (RAG) and IBM Granite that retrieves information from official admission documents and answers student queries in natural language with high accuracy.

The system should understand user questions, retrieve the most relevant content from institutional documents, and generate reliable responses using IBM Granite.

Proposed Solution

Our solution, **AdmitAI**, is an AI-powered Multi-College Admission Assistant built using IBM Granite and Retrieval-Augmented Generation (RAG).

The system indexes official admission brochures, fee structures, hostel documents, placement reports, eligibility guidelines, and admission policies from multiple colleges.

When a student asks a question, the system:

1. Understands the natural language query.
2. Searches the FAISS vector database.
3. Retrieves the most relevant document chunks.
4. Sends the retrieved context to IBM Granite.
5. Generates an accurate response with document references.

This ensures that every answer is based on official institutional documents instead of hallucinated AI responses.

IBM Technologies Used

- IBM Granite LLM
- IBM watsonx.ai
- IBM Cloud Lite
- Retrieval-Augmented Generation (RAG)

Other Technologies

- Python
- Flask
- LangChain
- FAISS
- HuggingFace Embeddings
- HTML
- CSS
- JavaScript